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| Tool Version      : Vivado v.2023.1 (win64) Build 3865809 Sun May  7 15:05:29 MDT
2023
| Date              : Wed Sep  9 13:21:08 2026
| Host              : bedouinoureddine running 64-bit major release (build 9200)
| Command           : report_power -file
C:/Vivado_Projects/Harness_Test_System_3/Harness_Test_System/power_2.txt -name
power_1
| Design            : FPGA_Top
| Device            : xc7a100tcsg324-1
| Design State      : routed
| Grade             : commercial
| Process           : typical
| Characterization   : Production
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```

Power Report

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### 1. Summary

|                          |              |  |
|--------------------------|--------------|--|
| +-----+                  |              |  |
| Total On-Chip Power (W)  | 0.208        |  |
| Design Power Budget (W)  | Unspecified* |  |
| Power Budget Margin (W)  | NA           |  |
| Dynamic (W)              | 0.110        |  |
| Device Static (W)        | 0.097        |  |
| Effective TJA (C/W)      | 4.6          |  |
| Max Ambient (C)          | 84.1         |  |
| Junction Temperature (C) | 25.9         |  |
| Confidence Level         | Low          |  |
| Setting File             | ---          |  |
| Simulation Activity File | ---          |  |
| Design Nets Matched      | NA           |  |

```

+-----+-----+
* Specify Design Power Budget using, set_operating_conditions -design_power_budget
<value in Watts>

```

## 1.1 On-Chip Components

| On-Chip      | Power (W) | Used | Available | Utilization (%) |
|--------------|-----------|------|-----------|-----------------|
| Clocks       | 0.001     | 5    | ---       | ---             |
| Slice Logic  | 0.001     | 2484 | ---       | ---             |
| LUT as Logic | <0.001    | 802  | 63400     | 1.26            |
| CARRY4       | <0.001    | 181  | 15850     | 1.14            |
| Register     | <0.001    | 1039 | 126800    | 0.82            |
| Others       | 0.000     | 9    | ---       | ---             |
| Signals      | 0.001     | 2505 | ---       | ---             |
| MMCM         | 0.106     | 1    | 6         | 16.67           |
| DSPs         | <0.001    | 2    | 240       | 0.83            |
| I/O          | <0.001    | 20   | 210       | 9.52            |
| Static Power | 0.097     |      |           |                 |
| Total        | 0.208     |      |           |                 |

## 1.2 Power Supply Summary

| Source      | Voltage (V) | Total (A) | Dynamic (A) | Static (A) | Powerup (A) |
|-------------|-------------|-----------|-------------|------------|-------------|
| Budget (A)  | Margin (A)  |           |             |            |             |
| Vccint      | 1.000       | 0.020     | 0.004       | 0.015      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vccaux      | 1.800       | 0.077     | 0.059       | 0.018      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco33      | 3.300       | 0.004     | 0.000       | 0.004      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco25      | 2.500       | 0.000     | 0.000       | 0.000      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco18      | 1.800       | 0.000     | 0.000       | 0.000      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco15      | 1.500       | 0.000     | 0.000       | 0.000      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco135     | 1.350       | 0.000     | 0.000       | 0.000      | NA          |
| Unspecified | NA          |           |             |            |             |
| Vcco12      | 1.200       | 0.000     | 0.000       | 0.000      | NA          |

|                                                   |    |       |       |       |       |    |  |
|---------------------------------------------------|----|-------|-------|-------|-------|----|--|
| Unspecified                                       | NA |       |       |       |       |    |  |
| Vccaux_io                                         |    | 1.800 | 0.000 | 0.000 | 0.000 | NA |  |
| Unspecified                                       | NA |       |       |       |       |    |  |
| Vccbram                                           |    | 1.000 | 0.000 | 0.000 | 0.000 | NA |  |
| Unspecified                                       | NA |       |       |       |       |    |  |
| MGTAVcc                                           |    | 1.000 | 0.000 | 0.000 | 0.000 | NA |  |
| Unspecified                                       | NA |       |       |       |       |    |  |
| MGTAVtt                                           |    | 1.200 | 0.000 | 0.000 | 0.000 | NA |  |
| Unspecified                                       | NA |       |       |       |       |    |  |
| Vccadc                                            |    | 1.800 | 0.020 | 0.000 | 0.020 | NA |  |
| Unspecified                                       | NA |       |       |       |       |    |  |
| +-----+-----+-----+-----+-----+-----+-----+-----+ |    |       |       |       |       |    |  |
| -----+-----+                                      |    |       |       |       |       |    |  |

### 1.3 Confidence Level

|                             |                                                                                                            |                                        |
|-----------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------|
| -----                       |                                                                                                            |                                        |
| +-----+-----+-----+         |                                                                                                            |                                        |
| -----+                      |                                                                                                            |                                        |
| -----+                      |                                                                                                            |                                        |
| User Input Data             |                                                                                                            |                                        |
| Action                      |                                                                                                            |                                        |
| +-----+-----+               |                                                                                                            |                                        |
| -----+                      |                                                                                                            |                                        |
| -----+                      |                                                                                                            |                                        |
| Design implementation state | High                                                                                                       | Design is routed                       |
|                             |                                                                                                            |                                        |
| Clock nodes activity        | High                                                                                                       | User specified more than 95% of clocks |
|                             |                                                                                                            |                                        |
| I/O nodes activity          | Low                                                                                                        | More than 75% of inputs are missing    |
| user specification          | Provide missing input activity with simulation results or by editing the "By Resource Type -> I/Os" view   |                                        |
| Internal nodes activity     | Medium                                                                                                     | User specified less than 25% of        |
| internal nodes              | Provide missing internal nodes activity with simulation results or by editing the "By Resource Type" views |                                        |
| Device models               | High                                                                                                       | Device models are Production           |
|                             |                                                                                                            |                                        |
|                             |                                                                                                            |                                        |
|                             |                                                                                                            |                                        |
| Overall confidence level    | Low                                                                                                        |                                        |
|                             |                                                                                                            |                                        |
| +-----+-----+               |                                                                                                            |                                        |
| -----+                      |                                                                                                            |                                        |

## 2. Settings

### 2.1 Environment

|                       |                          |
|-----------------------|--------------------------|
| Ambient Temp (C)      | 25.0                     |
| ThetaJA (C/W)         | 4.6                      |
| Airflow (LFM)         | 250                      |
| Heat Sink             | medium (Medium Profile)  |
| ThetaSA (C/W)         | 4.6                      |
| Board Selection       | medium (10"x10")         |
| # of Board Layers     | 12to15 (12 to 15 Layers) |
| Board Temperature (C) | 25.0                     |

### 2.2 Clock Constraints

| Clock                | Domain                        | Constraint (ns) |
|----------------------|-------------------------------|-----------------|
| clk_out1_clk_wiz_0_1 | u_pll/inst/clk_out1_clk_wiz_0 | 25.0            |
| clkfbout_clk_wiz_0_1 | u_pll/inst/clkfbout_clk_wiz_0 | 10.0            |
| sys_clk_pin          | sys_clk                       | 10.0            |

## 3. Detailed Reports

### 3.1 By Hierarchy

| Name                  | Power (W) |
|-----------------------|-----------|
| FPGA_Top              | 0.110     |
| u_FPGA_Core           | 0.004     |
| u_Discrete_FIR_Filter | 0.003     |
| u_pll                 | 0.106     |
| inst                  | 0.106     |